[RESEARCH UPDATE]

LUXEMBOURG COMPANY LAW · BATCH 1

9 WORKS · 8,809 PASSAGES
15 TASKS · 101 CRITERIA

Research update: Opulent on Luxembourg company law

SUMMARY

Today we are sharing our first results on the corpus you gave us: a deliberately naive version of Opulent that outperforms Harvey's open-source legal agent harness on held-out Luxembourg company-law work.

OPULENT AI RESEARCH · PREPARED FOR FRANÇOIS BERNARD · SEPTEMBER 30, 2026

HELD-OUT COMPLETION

98%

GLM-5.3, naive Opulent harness

LEAD OVER HARVEY LAB

at least 9.9 pts

On every metric we report

YOUR NINE WORKS

8,809

Cited passages, each traceable

HELD-OUT TASKS

12

Client work products, graded on source text

In the eight days since your nine texts reached us, our research has pursued two aims:

01 Measuring legal work on Luxembourg law with evaluations a partner would recognize as real work; and

02 Establishing how much of an agent's legal performance comes from the harness around the model, as distinct from the model itself.

This note reports where that work stands, including our first head-to-head result against the harness Harvey publishes with its Legal Agent Bench (LAB). The model we used is of modest capability by current standards, and the results are strong on both accuracy and completion.

[01]

Results

Our goal was to measure one thing cleanly: with the model held fixed, does the harness change the quality of legal work? We ran both harnesses with GLM-5.3 on the most rigorous held-out set we built from your corpus, twelve pieces of client work product, each graded against atomic criteria anchored in the exact text of your nine works. Opulent reaches 98 percent completion on the held-out set and outperforms Harvey's harness by at least 9.9 percentage points on every metric we report (Figure 1). The margin appears at a modest level of model intelligence, and that is what makes it informative: when the model is the same, the difference between the two systems is the way the work is organized around it.

Held-out completion, and the earlier paired run FIGURE 1

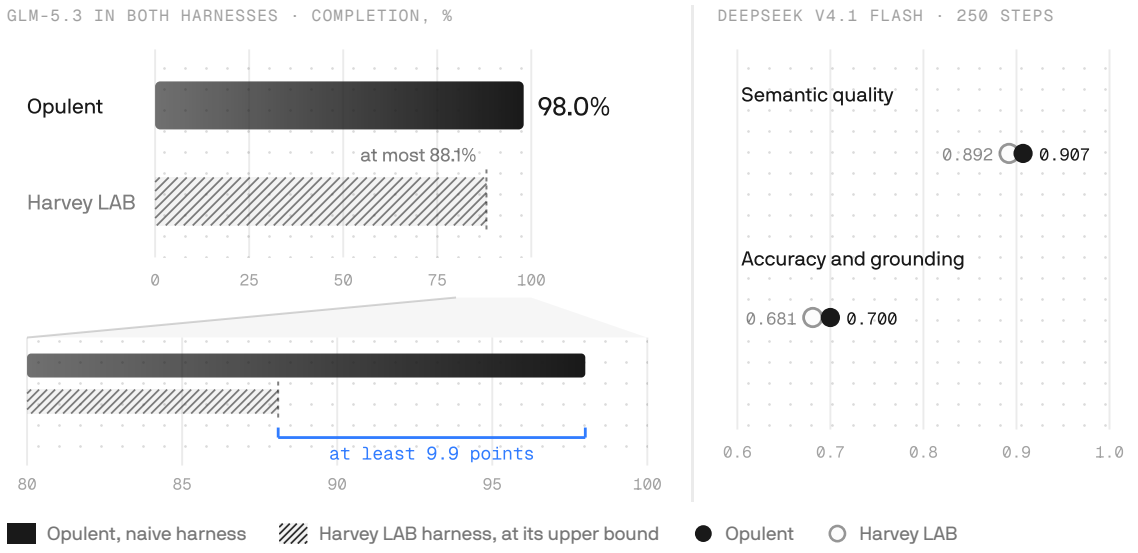


Figure 1. Left: completion on the twelve held-out work products, with GLM-5.3 answering in both harnesses. Harvey's bar is drawn at its upper bound, 9.9 points below Opulent, and the lower scale zooms in on 80 to 100. Right: the earlier 250-step run on the same tasks with DeepSeek V4.1 Flash, as trimmed means of Jev scores normalized to 0 to 1.

SOURCE · GLM-5.3 RUN, TEAM-REPORTED, ARTIFACTS PENDING COMMIT · EVAL/RLM-V5-COMPARISON-REPORT.MD

The result extends a pattern from our earlier runs, and widens it. On a 250-step run with DeepSeek V4.1 Flash over the same twelve tasks, Opulent led on semantic quality (0.907 against 0.892) and on accuracy and grounding (0.700 against 0.681), after removing each system's best and worst task. That report called the edge small, and it was. Our earlier question-answering comparisons on a fifty-question development set were close and went both ways, depending on the model and the judge. We keep every run, and list the principal ones in the appendix.

Opulent also performs well where the work is answering questions rather than drafting. On the fifty-question development set, graded against 178 atomic criteria, our agent passes 93.8 percent of criteria, and 48 of its 50 answers are grounded in

passages it actually read. That score was 73.0 percent before a single change to the harness, which we describe below.

Our second finding concerns ownership of the measurement. Every evaluation in this note is built from your own texts, graded against their exact wording, and frozen before any system answers. That makes it a fixed instrument: it can measure any system a firm is considering, including ours, on the law your practice actually uses, and it will measure the next model or harness the same way.

Full details on our methodology are in the appendix.

[02]

How we built it

Six pieces of work sit underneath these results: how we broke down your corpus, the evaluations and environments we built from it, the retrieval layer, the judge, the harness, and the head-to-head comparison. We describe each in turn.

02.1

Breaking down the corpus

Your nine works arrived as 78 MB of scanned and digital PDFs: the law of 10 August 1915 on commercial companies in its consolidation of 29 May 2026; the company-law treatises of Steichen, Winandy, and Payen and Mbayi; volumes on obligations, commandites, accounting law and the Belgian reform; and your own monograph on cross-border demergers. We extracted about 15.5 million characters of French text, using OCR for scanned pages, and parsed each work along its own structure: livres, titres, chapitres and sections for the treatises, and articles for the statute. The result is a typed knowledge graph of 4,781 structural nodes and 2,018 articles, divided into 8,809 passages (Figure 2). Each passage carries a stable citation to its work and to its place in that work, and each work is linked to its authors and to nine topics, from the primary statute to listed companies. An agent can therefore go straight to an article of the 1915 law, or to the passages of a treatise that discuss it, and cite exactly what it read.

02.2

Frontier evaluations and RL environments

We built two evaluations from the corpus, and packaged the harder one as reinforcement-learning environments. The first is a development set of fifty questions across nine kinds of legal question, from statute lookups and definitions to doctrine, procedure, comparative and multilingual questions, graded against 178 atomic criteria. The second follows the design of Harvey's LAB: fifteen pieces of client work product, each framed as a partner's instruction to an associate, with 38 fictional client files and 101 criteria anchored to 59 exact passages of your corpus (Figure 3). The tasks ask for deliverables a Luxembourg practice produces, such as the closing sequence for a complete Luxembourg–France split, an allocation ledger with creditor stress tests for a partial European split, vote tabulations for parallel capital actions, and a financing choice between a Belgian SRL and a

Luxembourg SARL. Separate model authors and reviewers checked every packet against the source text, and the suite is frozen with cryptographic hashes, so any later change to a task or rubric is detectable. Each packet has the three parts of a training environment: an instruction, a client matter placed in a sandbox, and a rubric whose score can serve as the reward. The same task that grades an agent today can train one tomorrow.

Your nine works, as the agent sees them

FIGURE 2

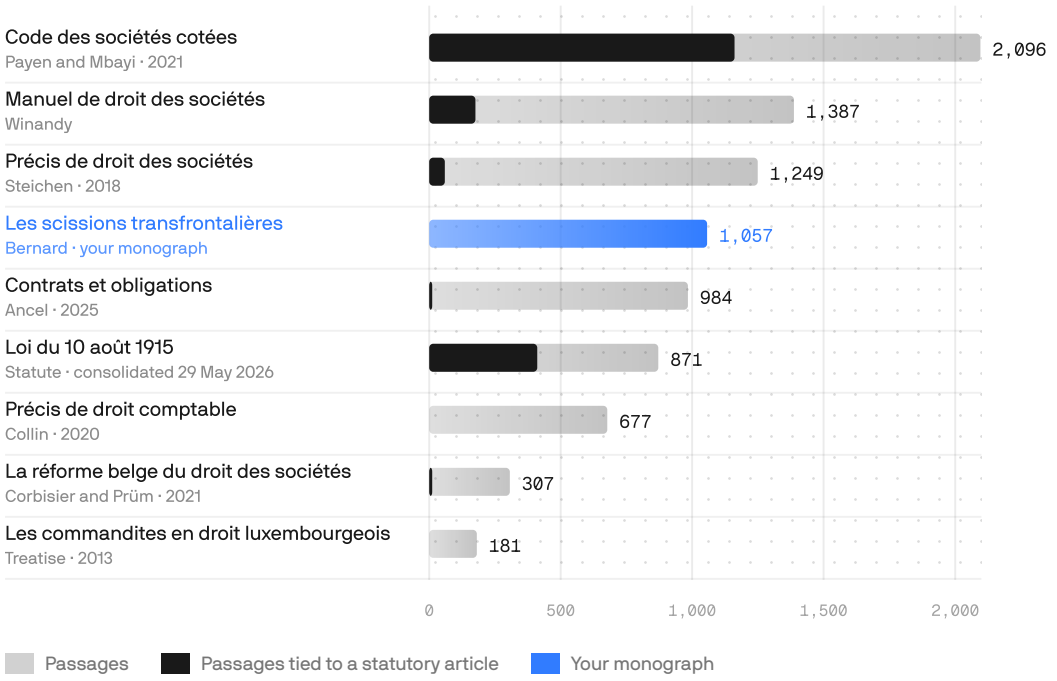


Figure 2. Passages per work, 8,809 in all. The darker segment marks the 1,814 passages tagged with a statutory article, concentrated in the 1915 law and the annotated code of listed companies. Works are credited by the names on their files.

SOURCE · CORPUS/INDEX/DOCUMENTS.CSV · CORPUS/INDEX/CHUNKS.JSONL

02.3

Embedding and retrieval layers

Retrieval decides what a model can know, so we built it across several frontier and emerging services and measured each configuration. Every passage is embedded with OpenAI's text-embedding-3-large at 3,072 dimensions and indexed in turbopuffer, beside a French full-text index that weights a work's title three times and its structural path twice as heavily as its body text. Vector and keyword results are fused, then reordered by a multilingual reranker. A second index uses Voyage 4 Large embeddings and Voyage's reranker, in namespaces of its own so that the two vector spaces never mix. Polygres, a Postgres service, holds a provenance mirror of every work and passage, so each citation joins back to its source record. On the development set, hybrid search with reranking returns the right passage first for 29 of 50 questions, against 20 for keyword search alone, raises mean reciprocal rank from 0.471 to 0.678, and places the right work in the top five for all 50 (Figure 4).

Two evaluations built from the corpus

FIGURE 3

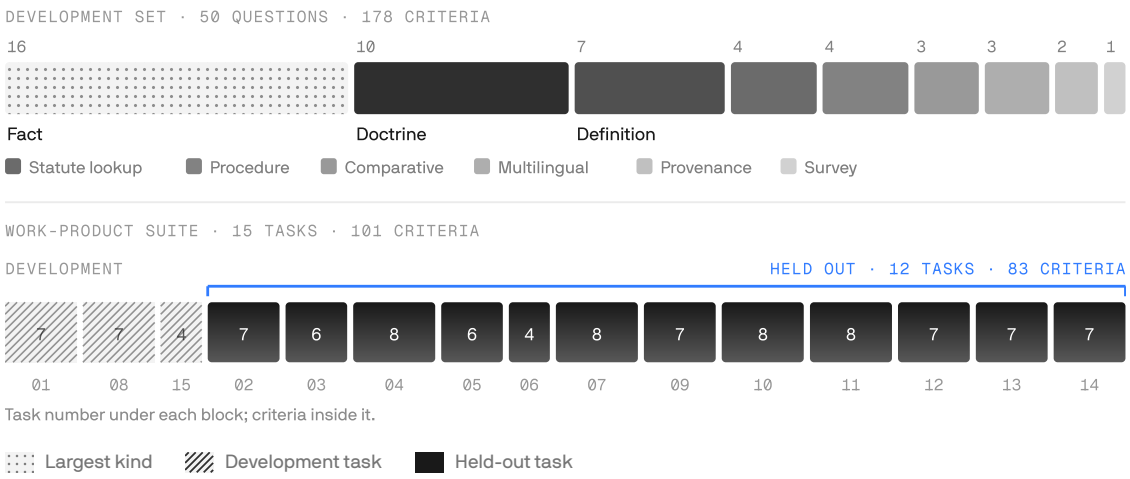


Figure 3. Top: the fifty development questions by kind, graded against 178 criteria. Bottom: the fifteen work-product tasks, each block sized by its criteria. The three development tasks are used for smoke tests and judge calibration; the twelve held-out tasks carry 83 of the 101 criteria.

SOURCE · EVAL/GOLD.JSON · EVAL/COMPLEX-V1/MANIFEST.JSON · EVAL/COMPLEX-V1/RUN-PLAN.MD

Retrieval on the development set

FIGURE 4

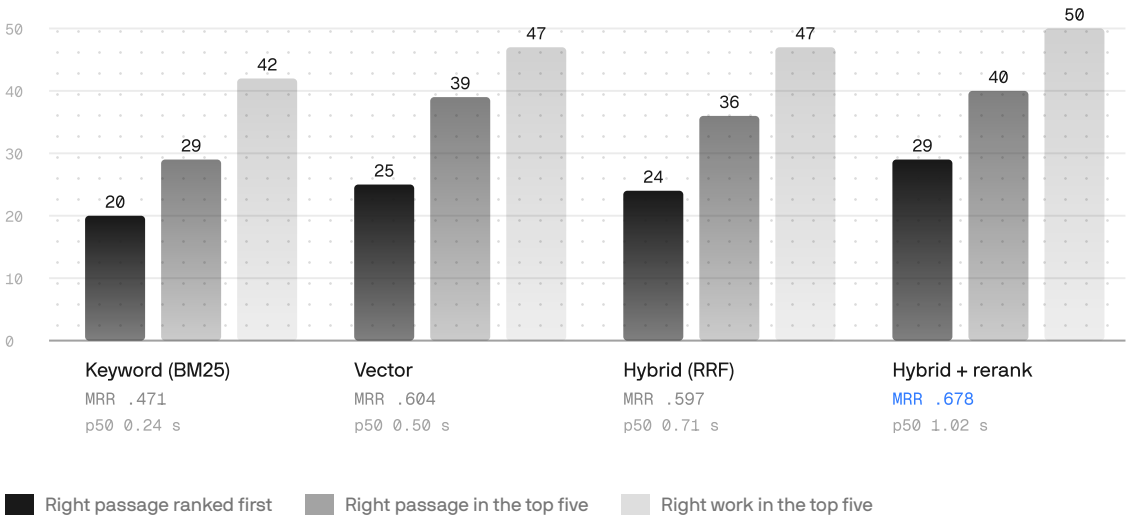


Figure 4. Questions out of 50 for each retrieval mode, with mean reciprocal rank (MRR) and median latency (p50). Hybrid search with reranking leads on every quality measure, at the cost of about three quarters of a second of latency over keyword search.

SOURCE · README.MD · EVAL/RESULTS.JSON

02.4

Jev as a System 1 judge

Most of our grading uses reasoning judges that deliberate before they rule. They are thorough and slow, and they can run out of room: in one smoke test, a high-reasoning judge spent 15,852 tokens reasoning about a single answer and returned a truncated verdict. We have been exploring where a System 1 judge, fast and intuitive in Kahneman's sense, fits beside them. Jev answers a narrowly

specified criterion in one pass and returns a score with a probability for each level of its scale, so it reports how sure it is along with what it concludes. In the 250-step comparison, Jev graded all 24 attempts with 1,224 output tokens in total, about 51 per attempt (Figure 5). That speed makes a new use practical: checking hyper-specific criteria, such as whether a vote tabulation uses the right denominator, instantly and on every draft, and passing only the uncertain cases to a slower judge or to a lawyer.

A System 1 judge beside the reasoning judges

FIGURE 5

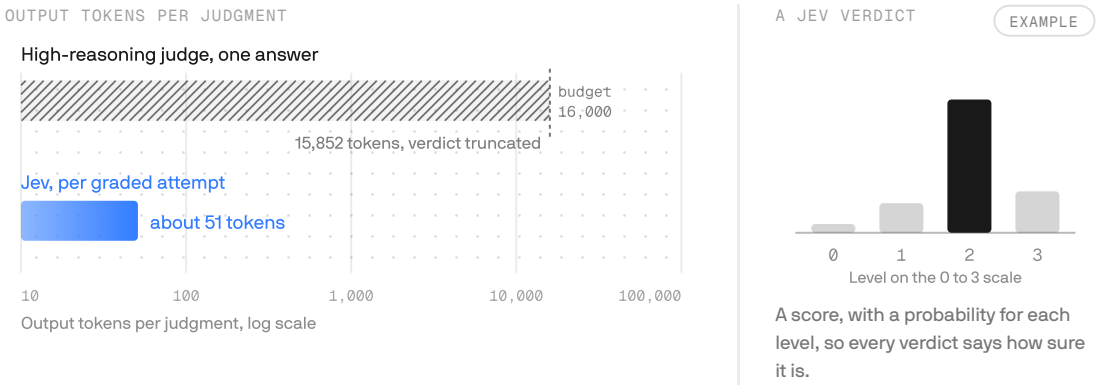


Figure 5. Left: output tokens per judgment, on a log scale. The reasoning judge's figure is one smoke test on a single answer, which exhausted its 16,000-token budget; Jev's is the mean over the 24 attempts of the 250-step run (1,224 tokens in all). Right: the shape of a Jev verdict, drawn for illustration without values.

SOURCE · EVAL/README.MD · EVAL/RLM-V5-COMPARISON-REPORT.MD

02.5

A naive cloud harness, built from scratch

To test the harness question directly, we built a naive version of Opulent from scratch in the cloud. A small control plane runs the model's turns, records every step and holds human approvals. Each agent works in its own E2B sandbox, work product is committed to code.storage, and the agent sees Opulent's tool surface: 97 tools, with the corpus reachable through search, passage and inventory tools served over MCP. The agent can keep a persistent Python session and hand parts of a task to sub-agents, each in its own sandbox, the recursive pattern that suits long legal materials. Everything else in the production product is left out, on purpose: if a plain version of the Opulent design outperforms Harvey's harness, the advantage belongs to the design, and the full product has more to add. Building it also showed how much small harness decisions matter. After we added one rule to the harness's instructions, to retry a filtered search that returns nothing without its filter before concluding that evidence is missing, the agent's score on the development set rose from 73.0 to 93.8 percent of criteria, and degraded runs fell from 18 to zero (Figure 6).

One harness rule, twenty points

FIGURE 6

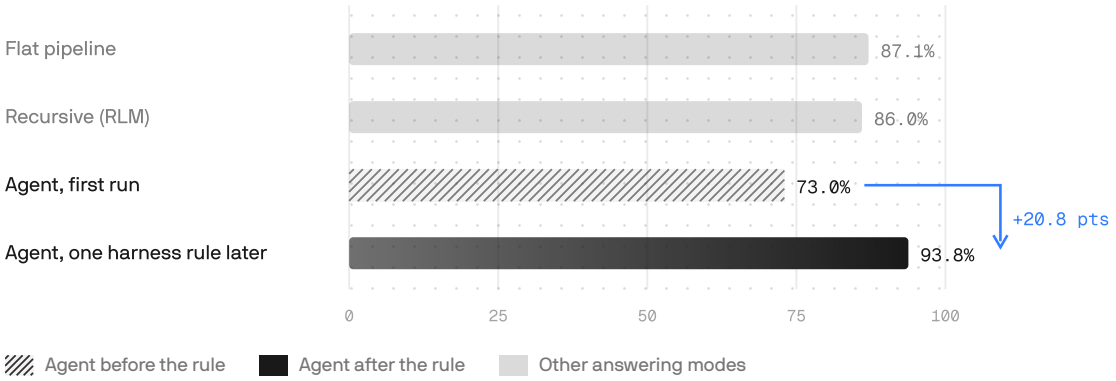


Figure 6. Share of the 178 development criteria passed by each answering mode. In its first run the agent pinned filters that matched nothing and concluded the evidence was insufficient. After one rule was added to the harness, to retry such a search without its filter, it passed 93.8 percent of criteria, above the fixed pipelines, and degraded runs fell from 18 to zero.

SOURCE · README.MD · EVAL/COMPOSE-AGENT-BASELINE.JSON · EVAL/COMPOSE-AGENT.JSON

02.6 Harvey's open-source harness, head to head

The two harnesses, on the same terms

FIGURE 7

	[A] HARVEY LAB HARNESS	[B] OPULENT, NAIVE HARNESS
Agent loop	LAB's own loop, adapter and prompts	A control plane with records and approvals
Tools	Six: bash, read, write, edit, glob, grep	Opulent's surface, 97 tools in the catalog
Corpus access	Exported passages, read as files	Search, passage and inventory tools over MCP
Sandboxes	E2B, in place of its Podman backend	E2B, one sandbox per agent
Delegation	A single agent	Sub-agents from a persistent Python session
HELD EQUAL	The model, the task files, the 8,809 passages, the step and token budgets, and the evidence rules	

Figure 7. What differs between the two arms, and what is held equal. We compare configured systems, so retrieval, prompts and tools differ along with the harness. The Harvey arm is Harvey's public LAB harness at a pinned commit.

SOURCE · EVAL/HARVEY-COMPARISON.MD · RUNTIME/README.MD · EVAL/COMPLEX-V1/RUN-PLAN.MD

For the comparison, we ran the harness Harvey publishes with LAB, the most complete open-source legal agent harness we know of: its own agent loop, prompts and six tools (bash, read, write, edit, glob and grep), pinned to a fixed public commit, with E2B sandboxes in place of its native container backend. Both harnesses receive identical task files and the same 8,809 passages of your corpus, run the same model under matched step and token budgets (Figure 7), and must save their work product to disk, where a missing file counts as a failure. We compare configured systems, so retrieval, prompts and tools differ along with the harness itself, and the reference is Harvey's public harness; Harvey's hosted

product was not part of the test. On those terms, the naive Opulent harness reaches 98 percent completion on the held-out set and leads by at least 9.9 points on every metric.

[03]

What's next

This is a first result on your law, and a strong one. It shows that a well-designed harness lifts legal work substantially even at modest model intelligence, and that your corpus supports evaluations demanding enough to show it.

Next, we want to:

01 Bring lawyers into the grading. Our rubrics were written and checked with AI assistance against your texts, and are not yet certified by a lawyer. Ten development questions are already flagged for a lawyer's adjudication, where the question leaves a company form, jurisdiction or scope open. Review by your team would turn a strong measurement into an authoritative one.

02 Train on the environments. The work-product packets are ready to serve as reinforcement-learning environments, and post-training a model on them is the direct test of whether gains from the harness and the model compound.

03 Extend the corpus and the tasks to more of your practice, and run the full Opulent product beside the naive harness, reporting cost and token use alongside quality.

Legal work on Luxembourg law is measurable, and much of the gain lives in the harness. We would welcome an hour with you to walk through the tasks, the rubrics and the results.

[04]

Acknowledgements

We owe this work first to François Bernard, who gave us the corpus, including his own monograph on cross-border demergers. We are grateful to Harvey for publishing LAB and its harness in the open, which made a fair comparison possible, and to the teams behind the services we built on, including E2B, turbopuffer, Voyage AI, Polygres, code.storage, Vercel's AI Gateway and the makers of Jev.

[05]

Appendix

Here we describe the evaluation design, retrieval, judging, harness configuration and paired runs in more detail.

05.1

Evaluation design

The development set contains 50 questions: 16 fact questions, 10 on doctrine, 7 definitions, 4 statute lookups, 4 on procedure, 3 comparative, 3 multilingual, 2 on provenance and 1 survey question, graded against 178 atomic criteria in total. The judge receives the exact source text the model read. We treat this set as a development benchmark and never as a held-out measure: it guided our work on retrieval and harness policy.

The work-product suite contains 15 tasks built on the task, material and deliverable-scoped criteria of Harvey's public LAB design; it imports none of Harvey's legal answers. Every task requires a written memo that includes its requested tables, schedules, calculations or replacement clauses, drawn from 38 fictional client files in Markdown and CSV. The private rubric holds semantic pass or fail rules, minimal-pass and near-miss examples, and exact references to canonical passages and scenario facts. Tasks 01, 08 and 15 are reserved for development, smoke tests and judge calibration. The remaining twelve form the held-out set, which carries 83 of the 101 criteria and is always reported separately (Table 1).

Review covered legal applicability, authority and date, arithmetic, scenario sufficiency, alternative valid interpretations and repeat penalties, and it led to material corrections: requiring the actual capital threshold for an ordinary SARL, allowing a properly supported board-interim route for available reserves, and fixing the chronology of pre-approval and future contingency in task 11, among others. The 2021 Belgian commentary is treated as secondary historical material and not as certified current Belgian law, and outcomes that depend on foreign implementation, valuations or future approvals remain conditional in the rubric.

A freeze step binds the public inputs, the private rubrics, the protocol and all 8,809 canonical passage bodies into one hashed manifest. Public staging copies only each task and its declared materials into the answering sandbox, 53 files in all, and no rubric.

TABLE 1

The twelve held-out work products

TASK	WORK PRODUCT	CRITERIA
02	Multi-entity closing exception and dependency checklist	7
03	Year-end dividend income cut-off and assertion workpaper	6
04	Listed remuneration report evidence and publication audit	8
05	Research memo authority audit and citation repair	6
06	Parallel capital actions: vote tabulation and board certification pack	4
07	Listed-SA remuneration vote and payment remediation	8
09	SA capital-increase resolution and board execution	7
10	Luxembourg–France complete split: executable closing sequence	8
11	Partial European split: allocation, universal transfer and creditor fallback	8
12	Commandite selection and limited-partner governance diligence	7
13	Financing choice: Belgian SRL versus Luxembourg SARL and SARL-S	7
14	Industry contributions and capital and rights structuring	7
	Held-out set	**83**

SOURCE · EVAL/COMPLEX-V1/MANIFEST.JSON

05.2

Retrieval

Passages are cut along headings and article boundaries, up to 4,600 characters (roughly 1,100 to 1,200 tokens) with a 200-character overlap. The keyword leg is a French BM25 index over the passage text, the work's title and the structural path, weighted 1, 3 and 2. The vector leg uses text-embedding-3-large at 3,072 dimensions. The two are fused by reciprocal rank fusion and reordered by jina-reranker-v2-base-multilingual. Filters are inferred from article and statute references, and weak results below a score of 0.45 trigger an expanded search. The Voyage profile uses voyage-4-large at 1,024 dimensions with separate query and document input types, and rerank-2.5, in its own turbopuffer namespaces. Table 2 gives the development-set results for each mode.

TABLE 2

Retrieval modes on the development set

MODE	EVIDENCE @1	EVIDENCE @5	DOCUMENT @5	MRR	P50
Keyword (BM25)	20 / 50	29 / 50	42 / 50	0.471	0.24 s
Vector	25 / 50	39 / 50	47 / 50	0.604	0.50 s
Hybrid (RRF)	24 / 50	36 / 50	47 / 50	0.597	0.71 s
Hybrid + rerank	29 / 50	40 / 50	50 / 50	0.678	1.02 s

SOURCE · README.MD · EVAL/RESULTS.JSON

05.3 Judging

Every judge sees the evidence the model actually saw. A full passage reaches the judge only when its complete canonical text and citation appeared in the model's visible tool output; partial excerpts stay in the audit record and are never promoted to full sources. This can under-credit an answer built from partial excerpts, so we inspect those cases before attributing a gap to legal reasoning.

Our reasoning judges were GPT-6 Luna and DeepSeek V4.1 Flash at high reasoning effort, with budgets of up to 16,000 output tokens, under versioned policies. Saved answers can be replayed under a new judge without being regenerated, so a score change from a replay is always a grading change and never an answering improvement. Jev (typesafe-ai/jev, through Vercel's AI Gateway) scores each attempt on a four-level scale from 0 to 3, which we normalize to 0 to 1 by dividing by 3. In the 250-step comparison, we report trimmed means that drop each system's highest and lowest task for each metric.

05.4 Harness configuration

The Opulent arm is a Node 24 control plane with SQLite records, E2B sandboxes for execution and desktop work, and code.storage for committed work product. Its catalog lists 97 tools and 13 explicit exclusions, including the voice, wiki and settings surfaces. The corpus tools are served by a native MCP server and verified live before each run. A persistent Python kernel and a spawn tool give the agent recursive sub-agents, each in its own sandbox, with default limits of 64 tool calls per cell and 16 descendants.

The Harvey arm uses Harvey's LAB agent loop, model adapter, prompts and six-tool executor from a fork pinned to commit 845a0884, with an E2B backend in place of its native Podman backend. Both arms run with source documents write-protected and network access restricted, and both receive the corpus as exported passages with stable citations, without questions, gold answers, rubrics or credentials.

05.5 Comparison protocol

We declared the comparison protocol before the first work-product run. Later revisions, such as raising the completion-token budget, applied to both sides at once and are recorded as revised-protocol results. Both harnesses answer from identical task files and the same corpus export, with matched step and completion-token budgets and the same answering model. Each keeps its own tools and instructions, and the task-specific requirement to save a written deliverable is given to both in the same words. Saved deliverables, transcripts, tool results, the evidence each model actually saw, elapsed time, tool counts and reported token use are collected before any sandbox is torn down. A missing deliverable is an execution failure and is never reconstructed from the final chat message. Recoverable tool errors stay visible without stopping the run. The protocol also rules out choosing favorable tasks, suppressing failed runs, changing budgets for one side, or tuning a judge toward a preferred winner, and it

requires us to report where Opulent wins, ties or loses. Request timeouts and retries remain specific to each harness; we report that asymmetry and do not claim identical resource envelopes.

05.6

Paired runs

TABLE 3

The principal paired runs, Opulent against Harvey's LAB harness

RUN	EVALUATION	ANSWERING MODEL	OPULENT	HARVEY LAB
This note	12 held-out work products	GLM-5.3	98% completion	At least 9.9 points lower on every metric
250 steps, Jev judge	12 held-out work products	DeepSeek V4.1 Flash	0.907 semantic, 0.700 accuracy	0.892 semantic, 0.681 accuracy
Audited criteria	15 development questions	DeepSeek V4.1 Flash	48 of 50 criteria, 12 strict all-pass	48 of 50 criteria, 9 strict all-pass
Instruction-aligned	10 development questions	GPT-6 Luna	32 of 36 criteria	28 of 36 criteria
Legacy judge	10 development questions	DeepSeek V4.1 Flash	27 of 36 criteria	31 of 36 criteria

SOURCE · EVAL/RLM-V5-COMPARISON-REPORT.MD · EVAL/PREPARED-MODAL-FIFTEEN-REPORT.MD · EVAL/INSTRUCTION-ALIGNED-COMPARISON.MD · EVAL/MODAL-DEEPSEEK-TEN-REPORT.MD

In the ten-question run with the legacy judge, the judge was not given search excerpts that Opulent had actually seen, which confounds Harvey's lead there. Development-set runs overlap, so we never add them together.

05.7

Limitations

01 Results for the GLM-5.3 run are reported as our team recorded them; its artifacts are being added to the repository with the same provenance as the earlier runs.

02 Twelve paired tasks are a small sample. They describe performance on these tasks and are not proof of general superiority.

03 Held out means the twelve tasks are reported separately from development. Earlier runs on these tasks informed later revisions of the protocol, so results after the first run are revised-protocol results.

04 The rubrics are AI-assisted and source-reviewed, and not yet certified by a lawyer.

05 The comparison is against Harvey's public reference harness at a pinned commit. It says nothing about Harvey's hosted product.

06 In earlier runs Opulent used more tokens and time than Harvey's harness: in the 250-step run, 11.5 percent more answering tokens and a 54 percent higher median latency. We will report cost and latency for the GLM-5.3 run separately from quality.